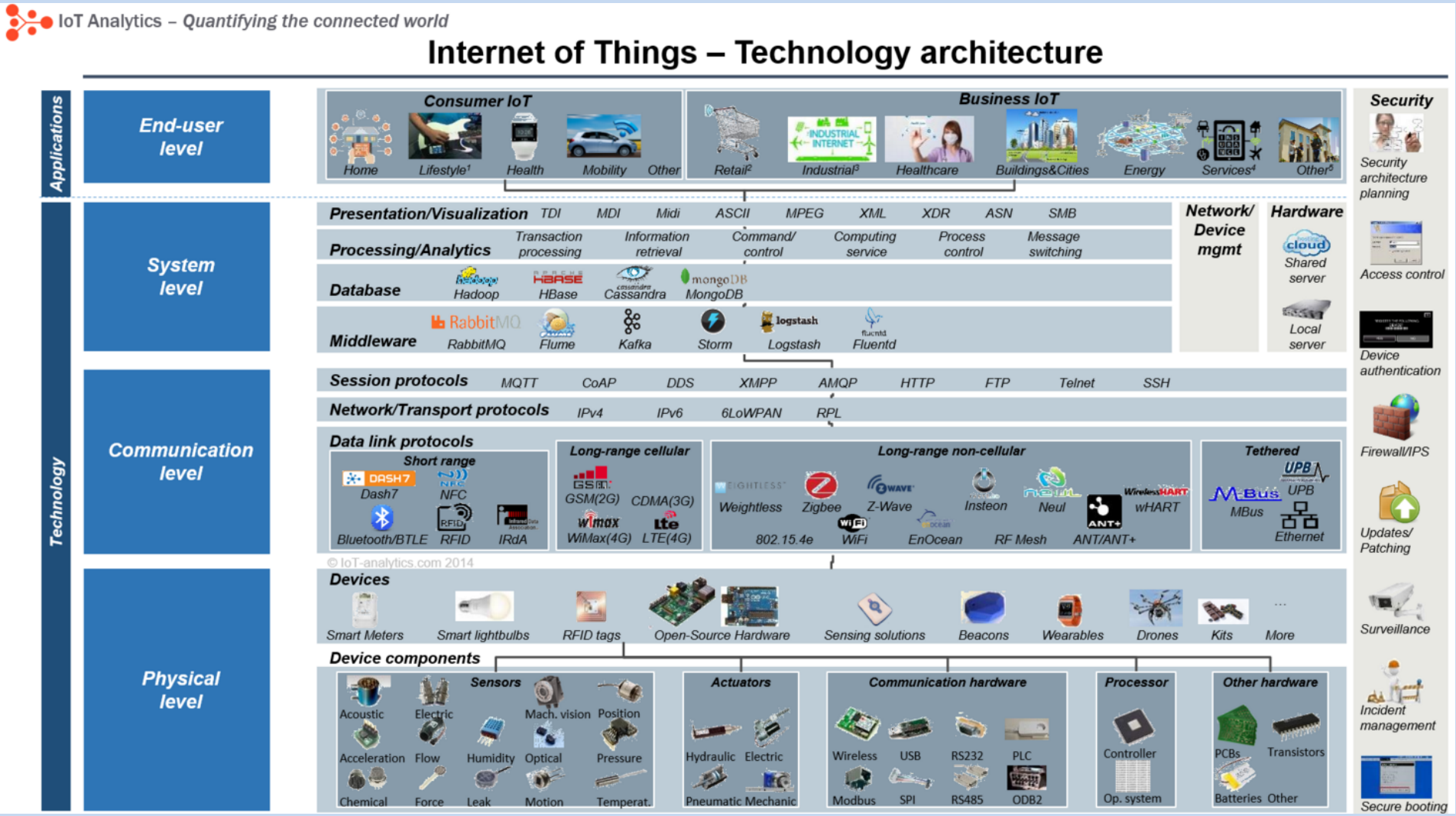
SAKARYA ÜNİVERSİTESİ
Bilgisayar ve Bilişim Bilimleri Fakültesi
Bilgisayar Mühendisliği Bölümü

BSM 451
INTERNET OF THINGS AND APPLICATIONS

IoT Wireless Communication Technologies RFID and NFC

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IoT Architecute

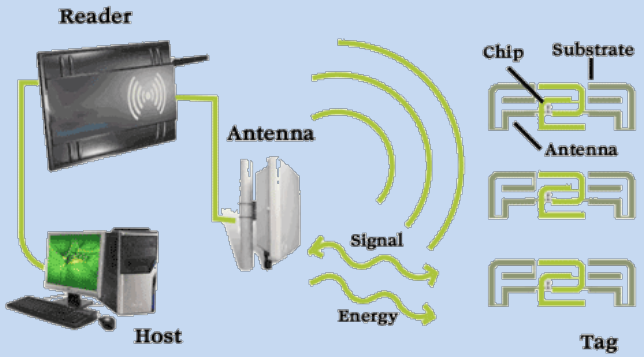


IoT Wireless Communication Technologies

- Short Range
  - **Radio-Frequency Identification, RFID,**
  - **Near Field Communications, NFC,**
  - Bluetooth Low Energy (BLE),
  - Infrared (IRdA)
- Non-Cellular Long-Range
  - Wireless Sensor Networks (IEEE 802.15.4 - ZigBee),
  - Z-Wave
  - ANT/ANT+
- Cellular Long-Range
  - GSM / GPRS
  - 3G / 4G (LTE) / 4.5G (LTE Advanced)
  - WiMAX

Radio-Frequency Identification (RFID)

- RFID (Radio Frequency IDentification): communication technology to transfer ID numbers of things and live entities over RF waves.
- RFID systems were first used in 1940s to identify airplanes (friend/enemy) in UK.
- Commercial applications appeared in 1990s.
- It is commonly used due to its features such contact-free, data read/write, and various transfer range.
- Provides security through cryptographic data transfer.

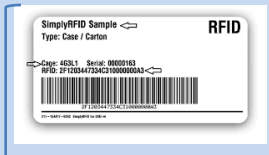


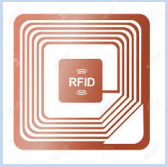
RFID System and Components

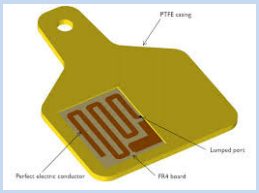
Active
RFID Tag



Passive
RFID Tag











RFID Antenna

RFID
Reader



RS232
RS485
RJ45

Middleware



Computer

RFID Tag

- A mobile object with limited memory.
- RFID tags can store not only ID numbers but also data such as manufacturing time, expiration date, etc.
- An RFID tag consists of 3 parts: microchip, antenna (coil) and substrate.





RFID Reader

- The fundamental purpose of RFID reader is to read ID numbers of RFID tags within the coverage area.
- In order to this, they provide energy required by passive tags to communicate by transmitting time-variable magnetic field.
- Reader transmit this magnetic field to tag with its antenna.





The data transfer between tag and the reader

- Data transfer between RFID reader and tag is provided by electromagnetic waves.
- Tag becomes active when it enters the electromagnetic field transmitted from the reader.
- The electromagnetic waves provide energy to the tag (by induced voltage on tag's coil) and data transfer from tag to reader is done by this energy. Reader converts data from tag to digital and transfers it to computer/server systems.

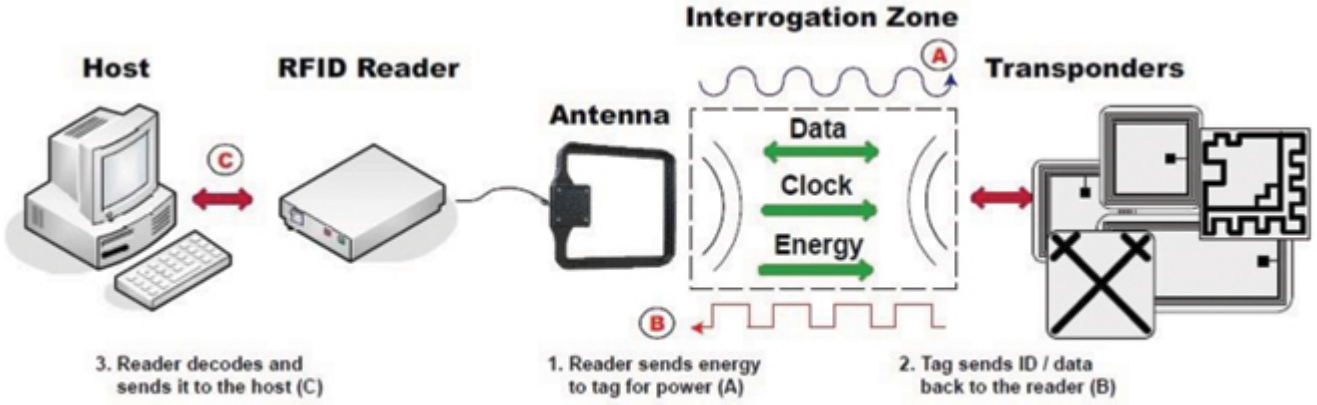


Figure 1 - Physical Architecture of RFID (GLOVER & BHATT, 2007)

This steps are for passive tags. Active tags have their own batteries.

RFID Tag Types

- By functions:
  - **Smart Tags:** They use batteries to run and keep their info. Long range and high price.
  - **Semi-Active Tags:** Microchip in tag is run by built-in battery. Energy required for data transfer is provided from electromagnetic field.
  - **Passive Tags:** No built-in power source. Cheap. Acts as a reflector. Gets energy in electromagnetic field and transfer data with this energy.

Active vs. Passive Tag





Feature	Passive tag	Active tag
Power source requirement	No	Yes
Energy	From reader's electromagnetic field	From their own power source
Reading range (depends on antenna)	3 – 5 m	Max. 300 m
Maintenance	No	Yes
Memory type	Mostly read	Mostly read and write
Cost	Cheap	Expensive
Size	Small	Large

RFID Tag Types

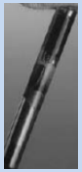
- By frequency:
  - Low Frequency, LF: 30 – 300 KHz. Range 1 – 5 cm.
  - High Frequency, HF: 3 – 30 MHz . Range 1 cm – 0.7 m.
  - Very High Frequency, VHF: 30 – 300 MHz . Range 1 – 3 m.
  - Ultra High Frequency, UHF: 300 – 1000 MHz . Range 1 – 3 m.
  - Microwave: 1 GHz and over. Range 1 – 10 m.

Note that usually tags which uses 13.56 MHz frequency band, are commercially used..

- By memory:
  - **Only readable tags:** Low capacity. Usually passive. Data stored while manufacturing and never changes.
  - **Readable and writeable tags:** High capacity. Can be passive or active. New info can be loaded by reader or current data can be altered. More costly.
  - **Write once read many (WORM):** writeable once during manufacturing, readable while using

- By physical shape:
  - Circular (disc),
  - Smart card,
  - Glass,
  - Watch,
  - Label,
  - Key chain,



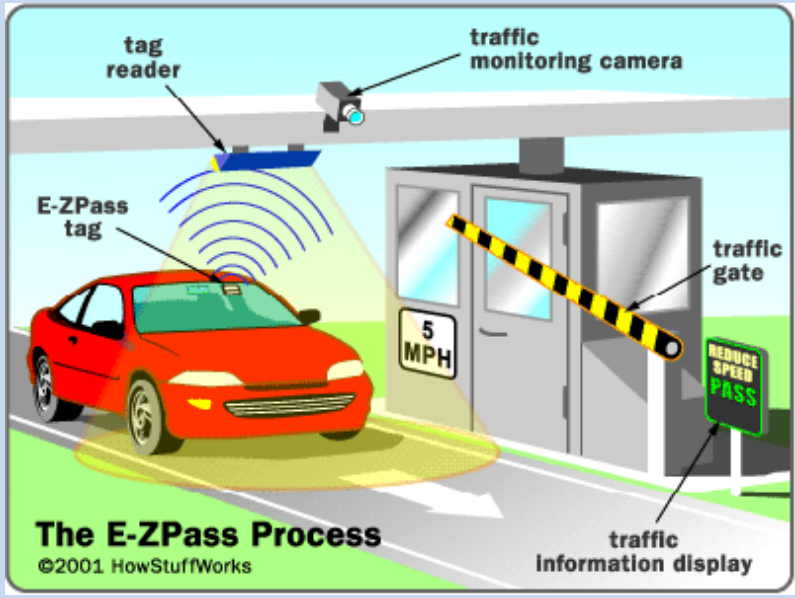


RFID Frequencies

Frequency band	Frequency Range	Region	Application
Low (LF)	120 – 135 KHz	USA, Canada, Japan, Europe	Short range, passive tags
High (HF)	13.56 MHz	USA, Canada, Japan, Europe	Smart cards and tags
Ultra High(UHF)	433 MHz	Most Europe, USA	Active low-power tags
	860 – 960 MHz	865- 868 MHz Europe 866-869 MHz S. Korea 902-928 MHz USA 952-954 MHz Japan	Supply chain applications
Microwave	2.450 GHz	USA, Canada, Japan, Europe	Long range and high speed data transfer with active tags

RFID Applications

- Paid tolls and bridges.
- Premise (building, etc.) entrance/exit control
  - Security control,
  - Parking/garage, etc.
- Contact-free payment systems
  - Public transportation,
  - Game arcade etc.







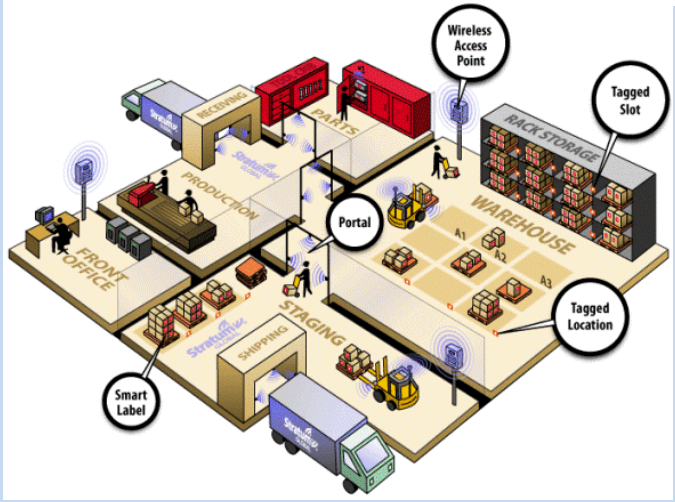


RFID Applications

- Library management system
  - Shelf management,
  - Book tracking (drop/return)
  - Protection against check out,
  - Stock monitoring.



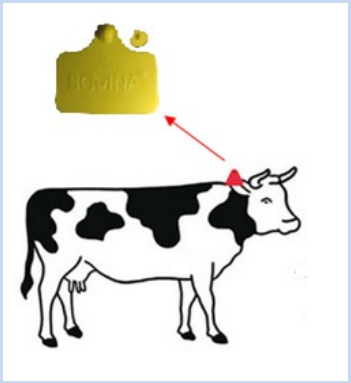
- Logistics
  - Distribution chain

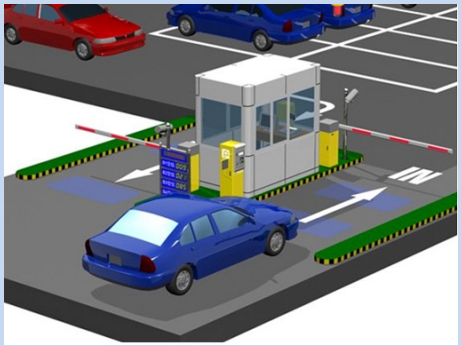




RFID Applications

- Identification applications (livestock, vehicle, human)

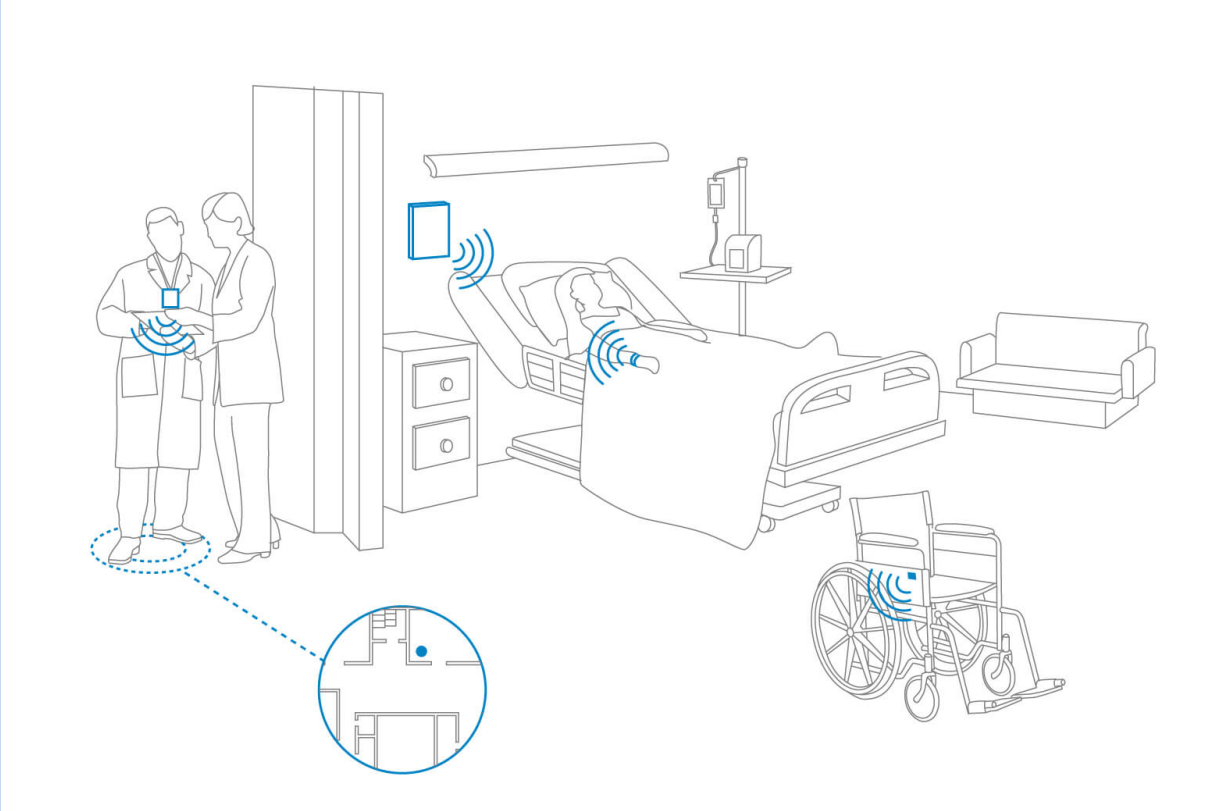






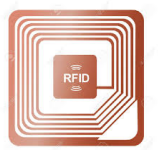
RFID Applications

- Health applications
  - Patient ID and treatment process,
  - Inventory/equipment monitoring,
  - Patient/employee tracking,



RFID vs. Barcode





Feature	Barcode	RFID
Tech	Optical	RF
Read/Write	Only readable	Read/Write
Memory	14 – 16 digits	96 – 256 digits
Storage (bytes)	1 - 100	16 – 64K
Line of sight requirement	Yes	No
Max. Range from reader	0 – 50 cm	0 – 5m
Rate	Low (~4s)	Very high (~0.5s)
Cost	Very cheap	10x barcode cost
Reliability	Sensitive	Reliable
Security	Low	High
Re-usable	No	Yes

Near Field Communication (NFC)

- NFC is short distance wireless communication technology based on high frequency low bandwidth RFID technology.
- Co-developed by Sony and Philips (NXP) in 2002.
- Standardized by ISO/IEC in 2003.
- Improved by NFC Forum founded in 2004.
  - Some important members of the forum: Samsung, Google, MasterCard, Visa, Microsoft, Texas Instruments.



NFC Technical Features

- Very short coverage area (<10 cm) (4 cm in practice),
- 13,56 MHz frequency,
- 424 Kbps data transmission rate,
- Low energy consumption,
- Short setup time,

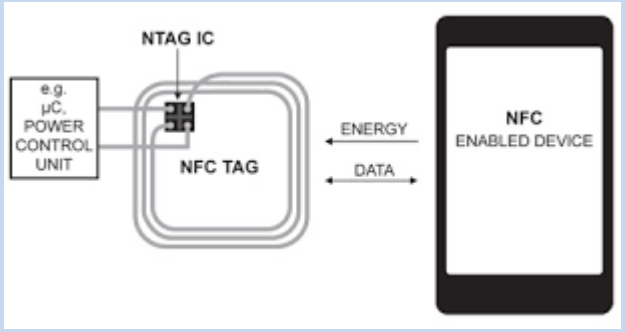


NFC Module and NFC Cards





- NFC systems usually include NFC tags and NFC tag readers.
- NFC Reader,
  - Devices that can read or write from NFC tags supported.



How does it work?







- **Read/Write Mode,**
  - Unidirectional data transfer mode. Used for active device (usually smart phone) to read or write data in NFC tag over another device.
- **Peer-to-Peer Mode,**
  - Data transfer between two devices. A device is active when transmitting and the other device is passive when receiving.
  - File transfer, social network applications, etc.
- **Card Emulation Mode,**
  - Designed for smart phones used for standard NFC smart card.
  - This secure mode can be used for mobile payment and public transportation.

NFC Applications

- Identification applications (livestock, vehicle, human)
- Contact-free payment,
- Health applications,
- Logistics,
- Library management,



NFC vs. RFID

- Developed on same standards..
- Setup time is same.
- Transmission range of NFC (10 cm) is shorter than RFID's. (3 m).
- NFC is more secure considering short range.
- NFC allows communication between two devices, a feature that does not exist in RFID.

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